

Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Status of the blind analysis and a request to open the control regions

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$\nu_\mu + \bar{\nu}_\mu$ CC single charged pion on argon • FHC, RHC and combined • blind, fake-data results

Internal note: [MicroBooNE-doc-46987](#)

Signal definition

Signal (truth level, charge-blind): $\nu_\mu/\bar{\nu}_\mu$ CC in the FV with exactly one μ , exactly one $\pi^\pm$, no π^0 , no kaons or other mesons, any number of nucleons; coherent production included.

Measured phase space: $p_\mu > 0.15 \text{ GeV}/c$, $p_\pi > 0.175 \text{ GeV}/c$, $\theta_{\mu\pi} < 2.6 \text{ rad}$

eff'turnon'pmu'fhc
(plot in production)

eff'turnon'ppi'fhc
(plot in production)

eff'turnon'thmu'fhc
(plot in production)

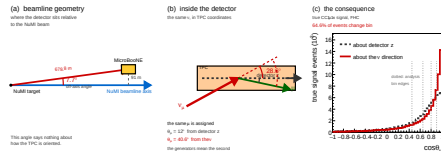
FHC selection efficiency for signal *without* the kinematic thresholds, each with the other two cuts applied; shaded: excluded. Momenta: efficiency turns on at the 10 cm muon and 20 cm pion track-length cuts; below threshold the correction would be large and model dependent. $\theta_{\mu\pi}$: set by purity, not efficiency; the region above 2.6 rad has 4.6% signal purity (cosmic dominated) and serves as the cosmic control region.

What is measured

Deliverables per configuration (FHC, RHC, combined)

- ▶ total flux-averaged cross section (one-bin extraction, no regularisation)
- ▶ $d\sigma/dp_\mu$, $d\sigma/d \cos \theta_\mu$ and $d\sigma/d\theta_\mu$, $d\sigma/d \cos \theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, two-bin $d\sigma/dp_\pi$
- ▶ proton-tagged subsample: integrated cross section (headline); TKI and W_{had} as secondary results

Blind analysis: every “data” point in this talk is Poisson-thrown central-value MC. No beam-on signal region has been opened.

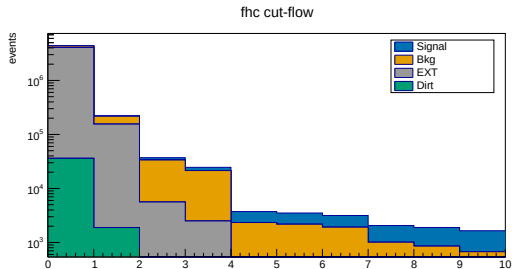


MicroBooNE sits 7.7° off the NuMI axis; the beam arrives 28° from detector z , so all angles and transverse quantities are computed in the beam frame.

Selection

- ▶ **Event:** software trigger, reconstructed vertex in the FV, topological score > 0.67
- ▶ **Muon:** highest muon-BDT score among qualifying tracks
- ▶ **Pion:** exactly one contained candidate (LLR, MIP-BDT, pion-BDT, Bragg- π), hits on all three planes
- ▶ **Topology:** shower veto, $\theta_{\mu\pi} < 2.6$ rad, non-proton multiplicity < 4 , fewer than 5 tracks

Largest remaining background: CC0 π with a proton taken as the pion.



FHC cut-flow, per-run POT-scaled, full CV weight.

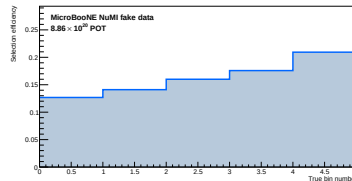
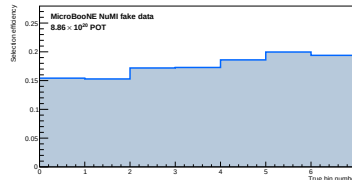
Yields and normalisation

Final cut	Signal	ν bkg	EXT	Pred.	Pur.
FHC	943	637	30	1 612	58.5%
RHC	1 015	720	23	1 759	57.7%
Comb.	1 959	1 357	53	3 372	58.1%

Efficiency 17.4–17.6% in the measured phase space.

Normalisation (details in backup)

- ▶ 8.86 (FHC) and 11.08×10^{20} POT (RHC); each run scaled by its own POT
- ▶ per-configuration new-G4 flux
- ▶ beam-off matched by run period and scaled by gates (1.9% of the sample)



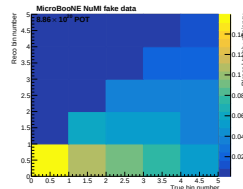
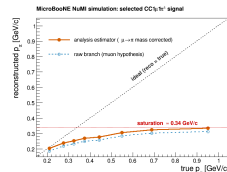
FHC selection efficiency vs. true p_μ and $\cos \theta_\mu$.

Binning requirements, and the pion momentum

Every binning must satisfy

1. bin width $\geq$ resolution; ≥ 45 selected signal events per bin
2. migration diagonal ≥ 0.68 in every true bin
3. unfolded: A_C well conditioned, closure passes, uncertainties not inflated

Observable	bins	min. diag. (FHC)	why not finer / coarser
$\cos \theta_\mu, \theta_\mu$	5, 5	0.68, 0.67	finer fails 2
$\cos \theta_\pi$	5	0.74	forward split; other splits fail 3
$\theta_{\mu\pi}$	5	0.79	finer passes 2, fails 3 ($s_{\min}/s_1 \rightarrow 0$)
p_μ	7	0.33	MCS 30%; kept, closure passes
p_π	2	0.91, 0.76	5 bins: 4 of 5 fail 2



Left: reconstructed p_π (range, mass-corrected) saturates near 0.34 GeV/c: pions interact before stopping. Right: five-bin p_π smearceptance (diagnostic).

- ▶ every estimator tried (golden pion, MCS, calorimetry, regression): a physics limit
- ▶ p_π released as **two bins**: [0.175, 0.205], > 0.205 GeV/c
- ▶ proton-tagged $W_{\pi p}$ shape **withdrawn**: no binning meets 1 and 2

Extraction: Wiener-SVD and what its integral means

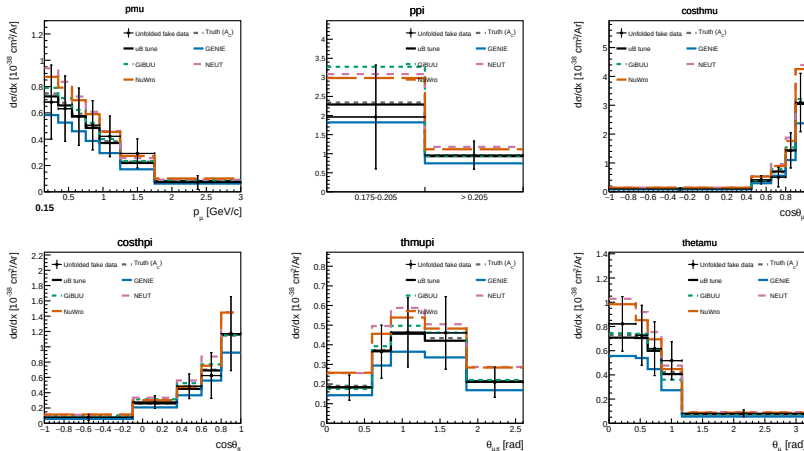
- ▶ $n^{\text{reco}} = S N^{\text{true}} + b$ with $S = \varepsilon M$ inverted once (no extra $1/\varepsilon$)
- ▶ Wiener-SVD (XSECANALYZER), second-derivative regularisation aware of the non-uniform bin widths
- ▶ the result is $A_C \times_{\text{true}}$; every prediction is smeared by the released A_C before comparison
- ▶ A_C is *not* norm preserving: the integral of a differential result is observable-dependent ($0.62\text{--}0.96 \times 10^{-38} \text{ cm}^2/\text{Ar}$); D'Agostini gives a flat, 23–31% higher integral
- ▶ the physical total is a one-bin extraction: full $\cos\theta_\mu$ range, $A_C \equiv 1$

Total flux-averaged cross section ($10^{-38} \text{ cm}^2/\text{Ar}$)

	inclusive	proton-tagged
FHC	1.03 ± 0.40	0.62 ± 0.26
RHC	0.93 ± 0.35	0.53 ± 0.25
Comb.	0.97 ± 0.37	0.57 ± 0.24

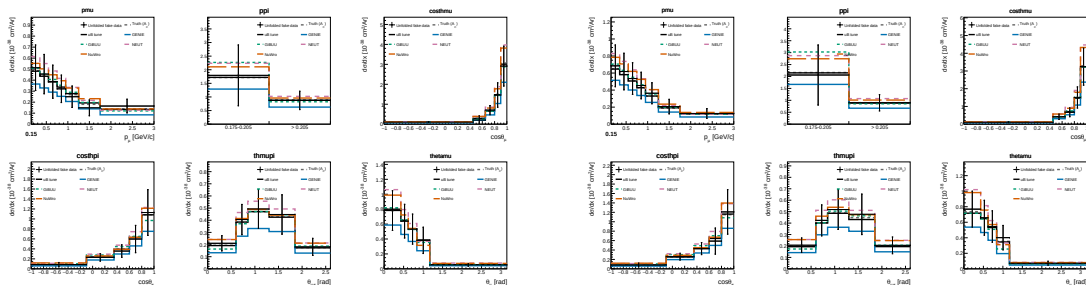
Extracted on the thrown fake data with the complete configured covariance; closes on the realised truth to 0.975–0.998. RHC/FHC of the tune = 0.90. Generators span $\times 1.6$, all within 1σ of the $\sim 38\%$ band: no discrimination at this exposure.

Differential results, FHC



Unfolded fake data, A_C -smeared truth, tune and the four generators, all in the A_C -smeared space. Generators differ by normalisation ($\times 1.6$, GENIE to NEUT); shapes agree within the uncertainties. Covariance: flux, cross section, re-interaction, detector, POT, targets, statistics.

RHC and combined

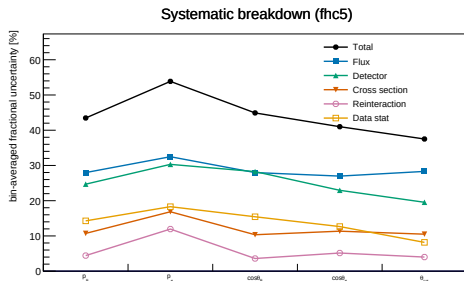


RHC

Combined FHC+RHC

Same 600 PPFX universes index-matched across both horn modes (rigorous flux correlation); native Run-4 RHC detector variations; each mode normalised to its own exposure and flux.

Uncertainties



Bin-averaged fractional uncertainty, inclusive (%):

Source	range	
Flux (PPFX)	24–33	dominant
Detector	16–35	~ flux for several obs.
Cross section	9–17	
Re-interaction	4–12	
Data statistics	6–18	
Total	32–54	

- ▶ the $\sim 17\%$ flux normalisation becomes $\sim 30\%$ through the correlated background subtraction, $17\% \times (1 + B/S)$; the one-bin total (30.2%, no regularisation) shows it exactly
- ▶ all 51 extractions carry the complete configured covariance including detector variations; A_C and covariances released as text

Validation, and what the review found

Validation

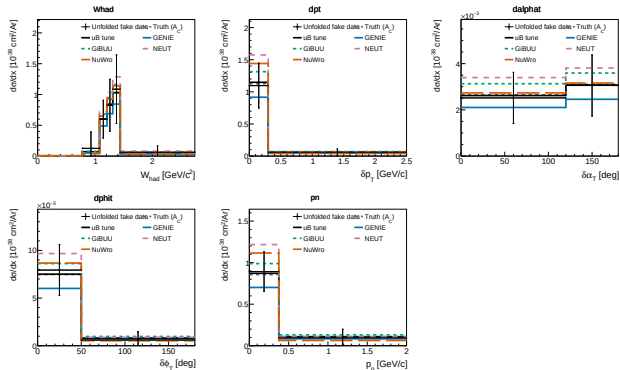
- ▶ fake-data closure: unfolded/realised truth = 1.010 on average over the 18 inclusive extractions; all $p > 0.9$
- ▶ 32-throw ensemble (current release): pull widths 1.08 (p_μ), 0.97 ($\cos\theta_\mu$), 1.02 (one-bin total); 62–71% / 92–94% coverage; means 0.5–2.7% low, at most 1.9σ , stated not corrected
- ▶ clean-cache rebuild of all 51 extractions bit-identical (2026-09-02 pass; to be repeated before unblinding)
- ▶ five automated guards: normalisation manifest, systematics/file coupling, staleness, table consistency, blinding guard

Corrected since August (all in the change log)

- ▶ beam-off scaled by an *event* count instead of gates: EXT was $9.7\times$ too large ($128 \rightarrow 13$ FHC events), then per run period ($13 \rightarrow 30$)
- ▶ fake-data truth carried the CV weight twice (1.6% on closure)
- ▶ PPFX CV weight missing from count-level tables (3–6%)
- ▶ one safe-weight rule everywhere; fake data re-thrown with it (1%)
- ▶ A_C applied twice to models; TKI computed about detector z (both withdrawn and redone)

None of these touched the real-data path; several would have biased a real-data result.

Proton-tagged subsample

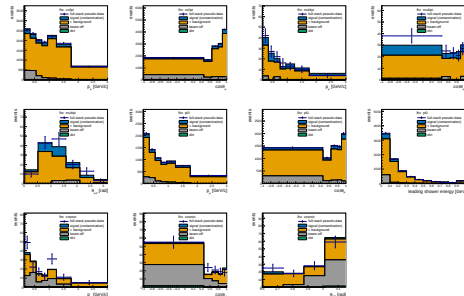


- ▶ inclusive selection + a contained proton $> 0.3 \text{ GeV}/c$; efficiency 11.4%, purity $\sim 50\%$; strong cosmic rejector
- ▶ TKI (δp_T , $\delta\alpha_T$, $\delta\phi_T$, p_n) in two bins, beam frame; W_{had} in six
- ▶ single-nucleon STV prescription applied to the πp system: p_n is a proxy (stated limitation)
- ▶ generators: normalisation spread only; no FSI discrimination at this exposure
- ▶ headline: one-bin total 0.62 ± 0.26 (FHC)

Control regions

Region	Inversion	Role; $T = N^{\text{SR}}/N^{\text{CR}}$ of the target class
CC0 π	$n_\pi = 0$	tests 0 π feed-in; $T = 0.03$, $\delta T/T = 31\text{--}34\%$ (detector-limited); target purity 72%
multi- π π^0	$n_\pi \geq 2$ shower fails veto	diagnostic only; $T = 2.5$ π^0 leakage; target CC+NC π^0 (analyser-adopted; ratification requested), purity 50–51%; $T = 0.02$, $\delta T/T = 17\text{--}19\%$ (MC-stat limited)
cosmic	$\theta_{\mu\pi} > 2.6 \text{ rad}$	beam-off norm. and shape; 34–41% EXT purity, no fitted correction

- ▶ disjoint from the signal region event by event; multi- π and π^0 overlap ($\sim 40\%$ of multi- π)
- ▶ pseudo-data/prediction 0.99–1.12 everywhere; full-stack test (MC+EXT+dirt) at unity within statistics
- ▶ per-region systematics incl. detector variations; transfer factors as correlated ratios (flux cancels to 2–3%)



Frozen distributions and binning, FHC, full-stack pseudo-data in place of beam-on points.

Beam-on processing: readiness, and an exposure gap

The control-region skim runs on real beam-on files

- ▶ all eight staged beam-on files processed through the signal-region-stripped skim: an event is written only if it fails the full signal selection *and* is in ≥ 1 control region
- ▶ each job re-read its own output and would have deleted it otherwise: marker present, **no** written event carrying the signal-region flag, every written event in a region — all eight passed
- ▶ the skim carries a marker the cross-section extraction refuses; **the signal region has not been opened**

Both productions behave. The Run-4/5 beam-on files carry a vestigial software-trigger flag $\equiv 0$ and no companion branch, exactly as the CRT-era beam-off. The binding guard leaves it unbound (fail-open); without it Runs 4–5 would have processed to zero events silently. Verified on the beam-on files before processing.

Beam-on exposure available

	runs staged	POT/ 10^{20}	of mode
FHC	1, 4, 5	7.589	85.7%
RHC	1, 3, 4	8.491	76.6%

No Run-2 beam-on file exists anywhere on disk — the run range 6999–14116 is absent. Run 2 is 1.27×10^{20} POT in FHC and 2.59×10^{20} in RHC: **19.4% of the total exposure.** Beam-off can stand Run 1 in for Run 2, because it is a cosmic-only sample matched by run period; beam-on cannot, it is the measurement.

Either those files are staged, or Run 2 leaves the exposure and the normalisation is regenerated. The per-run beam-on exposures and gate counts also need confirming against the trigger table: the ntuples carry no exposure record of their own, and Run 1 is split between the horn modes inside its run range.

What we ask

Open the four inclusive control regions under the frozen procedure; the signal region and the proton-tagged regions stay blind

Frozen before opening and unchanged since: selections at branch level, run-dependent normalisation, beam-off scaling, displayed observables and binning, covariance, goodness-of-fit tests and the permitted responses to a discrepancy.

What we ask now

1. **Run 2.** Do beam-on data exist upstream? If yes, stage them; if not, Run 2 leaves the exposure and the whole normalisation is regenerated.
2. **Exposure.** Confirm the per-run beam-on POT and gate counts against the trigger table, and the FHC/RHC split of Run 1 at run 6748, before the comparison is made.
3. Ratify the comparison procedure and the decision rule; ratify the π^0 definition and target class, *diagnostic-only* for this opening — no fitted correction is pre-authorised at 50–51% purity with a class settled after the purities were known.
4. Independent check of the skim; hashes of the frozen executable, plot list and protocol macro into the approval record.

How a discrepancy would be handled

- ▶ the frozen response tree treats a single run period's excursion as a reconstruction or exposure issue to be investigated, *not* a model correction, and forbids dropping a period without independent evidence of a defect
- ▶ a fitted normalisation is permitted only for $CC0\pi$, only on a passing shape test, and carries the transfer uncertainty and the leverage of the fixed components

Before the signal region (unchanged, none met here)

- ▶ PPFX-corrected flux constants confirmed; NuWro/GiBUU fake-data closure on the released chain; MCS muon-momentum term bounded or propagated
- ▶ documented treatment of any control-region discrepancy; clean-cache reproduction
- ▶ inverted pion-PID region for the $CC0\pi$ misidentification transfer; inverted proton-PID region before any proton-tagged result

Summary

- ▶ A blind, fully validated extraction of $\nu_\mu + \bar{\nu}_\mu$ CC1 $\pi^\pm$ cross sections on argon in the NuMI beam: total and six differential distributions in FHC, RHC and combined, with complete configured covariances and released A_C matrices.
- ▶ Two physics limits set the scope: the pion range saturates (two-bin p_π ; no $W_{\pi p}$ shape), and A_C is not norm preserving (the total is a one-bin extraction).
- ▶ Fake-data totals $1.03/0.93/0.97 \times 10^{-38} \text{ cm}^2/\text{Ar}$ with $\sim 38\%$ (37.6–38.7%) uncertainty, flux- and detector-limited.
- ▶ A sequence of normalisation and weighting defects was found by review and fixed; automated guards now stop each of them from recurring.
- ▶ **We ask to open the four inclusive control regions under the frozen procedure.** The skim that keeps the signal region blind runs on the real beam-on files, and all eight staged files pass its checks. Two exposure questions come first: **no Run-2 beam-on data exist** (19.4% of the exposure), and the per-run beam-on exposures need confirming against the trigger table.

Documents ([MicroBooNE-doc-46987](#)): analysis note, technical supplement and change log (Draft 1.5, release 2026-09-06, document revision 2026-09-07); data release in `report/data_release/`.

Backup

Backup: exposure and normalisation

Configuration	Runs	POT (10^{20})
FHC	1, 2, 4, 5	8.857
RHC	1, 2, 3, 4a–c	11.082
Combined	all (14 overlay files)	19.939

	FHC	RHC
Run 1	3.283	0.605
Run 2	1.268	2.591
Run 3	–	5.003
Run 4	2.075	2.883
Run 5	2.231	–

Beam-on POT per run, 10^{20} .

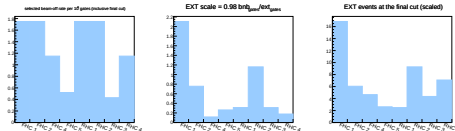
- ▶ every MC file scaled by $P_{\text{data}}(\text{run})/P_{\text{MC}}(\text{file})$; multi-file runs normalised by the sum of distinct file POT
- ▶ combined configuration = per-mode data exposures, one universe chain (600 PPFX universes index-matched across FHC and RHC)
- ▶ integrated flux (new-G4, $E_\nu > 60$ MeV): FHC 6.81, RHC 6.45, combined $6.61 \times 10^{-10} \text{ cm}^{-2}/\text{POT}$ (exposure-weighted); PPFX-corrected constants to be confirmed before unblinding
- ▶ $N_{\text{Ar}} = 8.71 \times 10^{29}$ in the fiducial volume (dead z slab removed)
- ▶ CV weight = tune $\times$ PPFX $\times$ normalisation; safe rule: non-finite, negative or $> 30 \rightarrow 1$, zero kept. GENIE universes carry the tune; PPFX universes are absolute; re-interaction universes relative

Backup: beam-off per run period

Mode	Run	Beam-on gates	EXT files	EXT gates	scale
FHC	1	9 846 635	run1	4 582 248	2.106
FHC	2	3 535 129	run1 (stand-in)	4 582 248	0.756
FHC	4	4 131 149	run4a–d	34 831 149	0.116
FHC	5	5 154 196	run5	19 256 341	0.262
RHC	1	1 458 253	run1	4 582 248	0.312
RHC	2	5 422 907	run1 (stand-in)	4 582 248	1.160
RHC	3	10 349 610	run3b	32 649 129	0.311
RHC	4	6 304 167	run4a–d	34 831 149	0.177

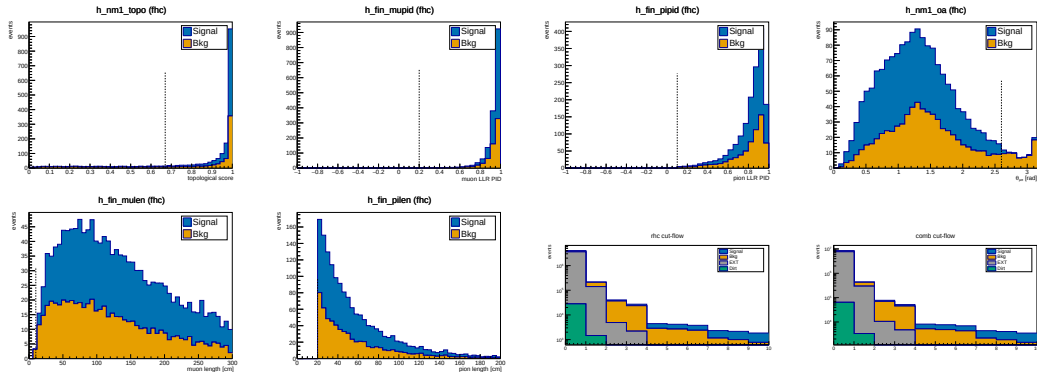
scale = $0.98 \times \text{beam-on gates} / \sum \text{EXT gates}$.

- ▶ cosmic-only data: the horn label is irrelevant, only the run period (CRT, detector conditions) matters
- ▶ no Run-2 beam-off exists; Run 1 is the stand-in (same pre-CRT conditions); Run 4 pools 4a–4d
- ▶ final-cut EXT: 30.2 (FHC), 23.2 (RHC), 53.4 (combined) events, 1.9% of the sample
- ▶ the previous release scaled by an *event* count: $9.7\times$ too much EXT



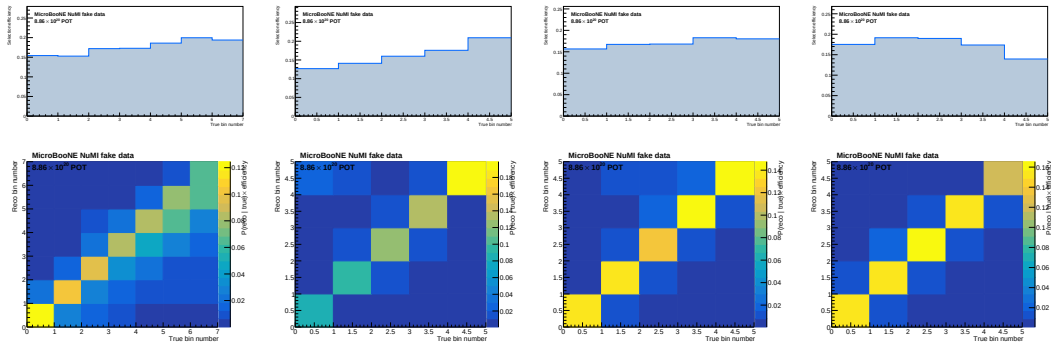
Selected cosmic rate per million gates: pre-CRT
Run 1 is $1.5\text{--}4\times$ the CRT-era runs (rates carry
16–35% Poisson errors).

Backup: selection variables (FHC)



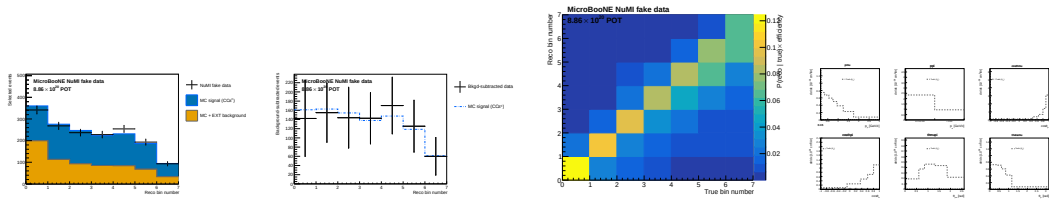
Top: topological score ($N - 1$), muon-BDT score, pion-BDT score, opening angle ($N - 1$). Bottom: muon and pion track lengths at the final cut; RHC and combined cut-flows. All per-run POT-scaled, full CV weight, fake data as points.

Backup: efficiency and response



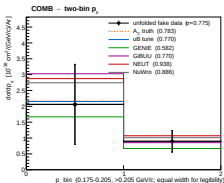
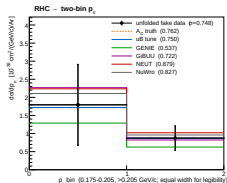
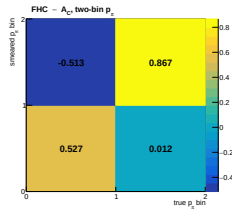
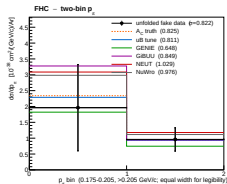
Top: selection efficiency vs. true variable (FHC). Bottom: smear acceptance $S = \varepsilon M$ used in $n^{\text{reco}} = S N^{\text{true}} + b$; top true bins are open-ended. Efficiency is folded into S once; no separate $1/\varepsilon$ division.

Backup: from reco distributions to the unfolded result (p_μ , FHC)



- ▶ left to right: reco-level stack with fake data; background-subtracted; the smearceptance S ; unfolded cross section with the A_C -smearred truth and models
- ▶ the Wiener-SVD regulariser is a second-derivative operator in bin index, corrected for the non-uniform bin widths (`fw_regmatrix`)
- ▶ the released products per extraction: A_C , unfolded values, the configured covariance, and the smeared predictions

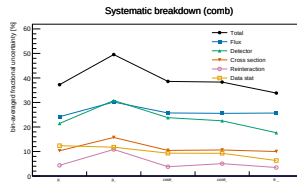
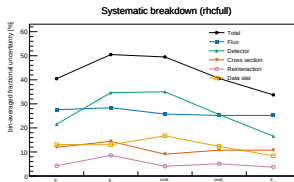
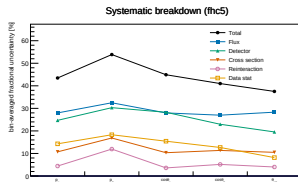
Backup: two-bin pion momentum



Two-bin $d\sigma/dp_\pi$ and its A_C : FHC (top), RHC and combined (bottom).

- ▶ range-based p_π under the muon hypothesis with the mass correction applied in the bin configuration; saturates at ~ 0.34 GeV/c because most pions interact before stopping
- ▶ five-bin diagnostic: top true bin only 6% diagonal (criterion > 0.68)
- ▶ estimators tried and rejected: golden-pion tagging ($40 \rightarrow 53\%$ golden at 87% efficiency, insufficient), MCS, calorimetric, regression on track variables, daughter recovery
- ▶ two bins $[0.175, 0.205]$ and > 0.205 GeV/c: diagonals 91% / 76%, released for all three configurations and the proton-tagged subsample
- ▶ same limit removes the proton-tagged $W_{\pi p}$ shape (pion momentum enters directly)

Backup: uncertainty breakdown by observable and configuration



Flux: 600 PPFX universes.

Cross section: GENIE multisim
+ unisims.

Re-interaction: Geant4
reweight.

Detector: Run-4 variation
samples, FHC and RHC na-
tively; proton-tagged included.
POT 2%, targets 1%, statistics
of MC and fake data.

Detector variations are 16–
35% after unfolding (inclusive;
10–109% proton-tagged).

Bin-averaged fractional uncertainty by source and observable, FHC, RHC and combined (2026-09-06 release).

Backup: closure and ensemble tests

Fake-data closure (18 inclusive extractions)

- ▶ fake data: an external Poisson throw of the CV prediction per run, weights reset to one in both reco and truth
- ▶ unfolded / realised truth = 1.010 on average; χ^2 probabilities > 0.9 everywhere
- ▶ no alternative-generator fake-data test exists on the released chain yet (a NuWro/GiBUU pass is a prerequisite for unblinding)
- ▶ the A_C -smeared integral is observable-dependent by construction; D'Agostini gives a flat integral 23–31% higher, iteration-stable (it returns its prior on CV input)

32-throw ensemble

- ▶ pull widths 1.08 (p_μ), 0.97 ($\cos\theta_\mu$) and 1.02 (one-bin total); 68% and 95% coverage recovered within the ensemble statistics
- ▶ ensemble means 0.5–2.7% below the reference (at most 1.9σ ; largest in the most regularised observable), stated and not corrected
- ▶ $\cos\theta_\mu$: the strongly suppressing A_C does not distort the uncertainties
- ▶ θ_μ and $\cos\theta_\mu$ are different partitions of one phase space; after the width-aware regularisation both discriminate, θ_μ with systematically larger χ^2 (model statements use $\cos\theta_\mu$)

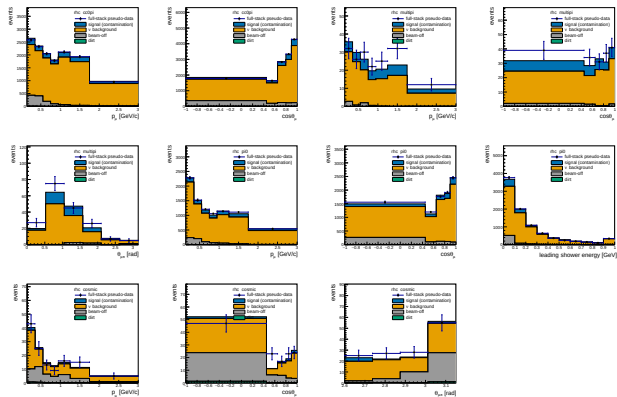
Reproducibility: clean-cache rebuild of all 51 extractions bit-identical on the 2026-09-02 pass, to be repeated on the release before unblinding; the universe cache is keyed on the normalisation scheme and on the file list.

Backup: control-region transfer factors (FHC)

Class	SR	CC0 π	multi- π	π^0	cosmic	$T_{\text{CC0}\pi}$	T_{multi}	T_{π^0}	T_{cosmic}
signal (contamination)	943	1 124	29.7	662	6.0	0.84	32	1.43	158
ν_μ CC 0 π (track taken as pion)	244	8 781	2.6	1 011	23.2	0.03	95	0.24	11
ν_μ CC 1 $\pi^\pm$ out of phase space	52.7	629	2.7	361	14.8	0.08	19	0.15	3.57
ν_μ CC with a π^0	58.0	408	20.2	3 070	2.5	0.14	2.87	0.02	23
ν_μ CC $\geq 2\pi^\pm$, no π^0	145	222	57.8	658	3.9	0.65	2.51	0.22	37
out of fiducial volume	47.4	480	3.3	253	18.5	0.10	14	0.19	2.56
NC and ν_e CC	89.5	549	7.1	632	6.4	0.16	13	0.14	14
total ν -MC	1 581	12 192	124	6 646	75.3				
CC+NC π^0 (<i>adopted target; ratification requested</i>)	64.5	444	21.8	3 331	3.2	0.15	2.96	0.019	20

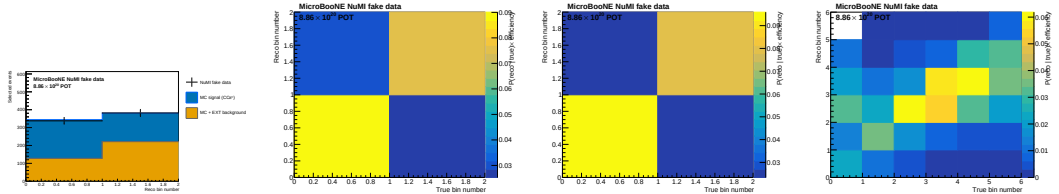
- ▶ $T = N^{\text{SR}}/N^{\text{CR}}$ per truth class; a normalisation change of the target class in a region propagates to the signal region by T
- ▶ CC0 π : $T = 0.03$, $\delta T/T = 31\text{--}34\%$ (genuine detector sensitivity); π^0 (adopted target CC+NC π^0 , ratification requested; purity 50–51%): $T = 0.02$, $\delta T/T = 17\text{--}19\%$, at its MC-statistical floor; flux cancels to 2–3% in the ratio
- ▶ the multi- π region has $T = 2.5$ and a 24% signal share: diagnostic only, no correction may be derived from it
- ▶ the cosmic region is 34–41% EXT with the rest dominated by CC0 π ; it tests the beam-off normalisation and shape, no fitted correction

Backup: control regions, RHC



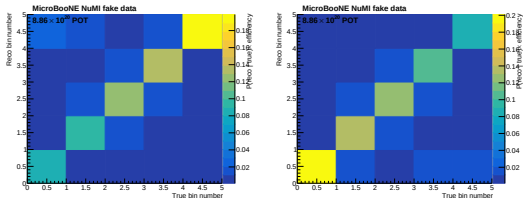
Frozen distributions (p_μ , $\cos\theta_\mu$, opening angle, shower energy) and binning for the four RHC regions; full-stack pseudo-data (MC+EXT+dirt, Poisson-thrown) in place of beam-on points. Pseudo-data/prediction 0.99–1.12.

Backup: proton-tagged kinematics



- ▶ TKI built in the NuMI beam frame from the μ and the leading proton with the pion added to the hadronic system; before the beam-frame fix δp_T was computed about detector z (0.74 vs. 0.26 GeV/ c mean)
- ▶ two-bin releases for δp_T , $\delta \alpha_T$, $\delta \phi_T$, p_n ; six bins in W_{had} ; $W_{\pi p}$ withdrawn (pion momentum)
- ▶ all proton-tagged extractions carry detector variations (10–109%, median 33%; largest for six-bin W_{had})
- ▶ the two-bin shapes separate generators by normalisation only; FSI discrimination would need the full exposure and better p_π

Backup: angles in the beam frame



Smearacceptance of $\cos\theta_\mu$ (left) and θ_μ (right), beam frame (drawn 2026-08-31; the response depends only on the simulation and the binning, unchanged since): two partitions of the same phase space (the θ_μ edges are not the arccos images of the $\cos\theta_\mu$ edges).

- ▶ beam direction: truth uses the per-event neutrino direction; reco uses target-to-vertex, with the convention of the NuMI ν_e analysis
- ▶ 64.6% of events changed $\cos\theta_\mu$ bin when the axis was corrected from detector z to the beam
- ▶ p_μ , p_π , $\theta_{\mu\pi}$ and the selection are frame-independent and were unaffected
- ▶ $\theta_{\mu\pi}$ coarsened $7 \rightarrow 5$ bins to limit A_C smearing; $\cos\theta_\pi$ re-binned to five forward-split bins $(-1, -0.1, 0.35, 0.6, 0.8, 1)$ after the 2026-09-05 re-check: better-conditioned A_C , same per-bin uncertainties